PROJECT 2 REPORT - EXPLORATORY DATA ANALYSIS (EDA)

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E-Commerce Sales Data Analysis

Industrial Training Kit | Batch: 2026 | DecodeLabs

1. Problem Statement

Core Question: What is the underlying “pulse” of our e-commerce business? This analysis investigates sales patterns, product performance, order health, and customer behavior to uncover actionable insights for strategic decision-making.

Key Objectives:

Identify revenue drivers and product performance

Assess order fulfillment health (statuses, cancellations, returns)

Detect market trends and seasonal patterns

Uncover anomalies and outliers that warrant investigation

Provide data-driven recommendations for business growth

2. Methodology

Data Source:

Dataset: 1,200 e-commerce orders

Time Period: January 2023 – June 2025

Attributes: 14 features including product details, pricing, payment methods, order status, and referral sources

Analytical Approach:

Data Validation: Verified integrity of the TotalPrice calculation (Quantity × UnitPrice = TotalPrice). Found 100% match, confirming data accuracy.

Descriptive Statistics: Calculated the five-number summary and measures of central tendency to understand the data’s “center of gravity.”

Univariate Analysis: Analyzed distributions of key variables (TotalPrice, Product, OrderStatus, PaymentMethod, ReferralSource).

Outlier Detection: Applied the IQR Method, which is robust against extreme values. Outliers were defined as values below Q1 − 1.5×IQR or above Q3 + 1.5×IQR.

Bivariate Analysis: Explored relationships between variables using PivotTables and cross-tabulations.

Trend Analysis: Examined temporal patterns by month.

3. Key Findings

3.1 The Geometry of Distribution: Total Price is Right-Skewed

Metric	Value
Minimum	$11.39
Q1 (25th Percentile)	$410.52
Median (50th Percentile)	$823.62
Q3 (75th Percentile)	$1,578.48
Maximum	$3,456.40
Mean	$1,053.97

Interpretation: The data is right-skewed (Mean > Median by $230.35). This indicates that while most orders cluster around moderate values ($410–$1,578), a minority of large, high-value orders pull the average upward.

3.2 Product Performance

Product	Total Revenue	Order Count
Chair	$195,620.11	562
Printer	$195,612.61	542
Laptop	$192,126.56	535
Tablet	$186,568.95	497
Monitor	$175,651.41	480
Desk	$167,459.93	508
Phone	$151,722.39	411
TOTAL	$1,264,761.96	3,535

Key Insight: Chair generates the highest revenue ($195,620.11), closely followed by Printer ($195,612.61). Phone has both the lowest revenue and the lowest unit volume among all products.

3.3 Order Health: The Fulfillment Fingerprint

Order Status	Count	Percentage
Cancelled	250	20.83%
Returned	247	20.58%
Shipped	235	19.58%
Pending	237	19.75%
Delivered	231	19.25%
TOTAL	1,200	100%

Red Flag: Only 38.83% of orders are successfully fulfilled (Delivered + Shipped). 41.42% are either Cancelled or Returned, a larger share than the orders that completed successfully.

3.4 Payment Method Analysis

Payment Method	Sum of TotalPrice
Cash	$259,786.29
Credit Card	$263,847.63
Debit Card	$232,361.18
Gift Card	$246,323.92
Online	$262,442.94

3.5 Referral Source Analysis

Referral Source	Sum of TotalPrice	Share of Revenue
Instagram	$275,285.45	21.76%
Email	$261,808.55	20.70%
Google	$250,441.48	19.80%
Facebook	$250,410.90	19.80%
Referral	$226,815.58	17.93%
TOTAL	$1,264,761.96	100%

Takeaway: Instagram is the strongest revenue driver, followed closely by Email. Facebook and Google are roughly tied. Direct Referral generates the least revenue of the five tracked sources.

3.6 Outlier Detection (IQR Method)

Q1 = $410.52   Q3 = $1,578.48   IQR = Q3 − Q1 = $1,167.96

Lower Bound = Q1 − 1.5×IQR = −$1,341.41 (effectively $0, since revenue cannot be negative)

Upper Bound = Q3 + 1.5×IQR = $3,330.41

Key Anomalies Identified:

OrderID	Product	Total Price	Concern
ORD200789	Tablet	$3,456.40	Above upper bound, highest value, potential bulk business
ORD201161	Phone	$11.39	Extreme low value, possible pricing error

Note: With the Upper Bound of $3,330.41, additional high-value orders close to this threshold should be re-screened, as the lower bound shift changes which orders qualify as outliers.

3.7 Monthly Revenue Trend

Month	Revenue
January	$124,313.23
February	$112,344.78
March	$123,840.93
April	$109,186.05
May	$135,142.59
June	$170,616.13
July	$85,784.64
August	$86,343.21

September	$69,321.65
October	$89,834.82
November	$75,493.43
December	$82,540.50

Best Month: June ($170,616.13). Revenue trends upward from January through a June peak, then drops sharply and stays consistently lower from July through December, with September the weakest month ($69,321.65). This split pattern (a stronger first half, a flatter and lower second half) is worth investigating, it may reflect seasonality, a change in marketing spend, or a data coverage gap in the second half of the year.

4. Key Performance Indicators (KPIs)

#	KPI	Value
1	Average Order Value (AOV)	$1,053.97
2	Total Revenue	$1,264,761.96
3	Total Orders	1,200
4	Order Fulfillment Rate	≈39% (38.83%)

5. Recommendations

Immediate Action Items (Next 30 Days)

Investigate High Cancellation/Return Rate (41.42% of orders, exceeding the 38.83% that are successfully fulfilled): conduct root-cause analysis, review shipping partners and product quality control, and target a reduction to under 20% within 6 months.

Investigate Extreme Outliers: review ORD201161 ($11.39 Phone) for a possible pricing error, and re-screen high-value orders against the corrected upper bound of $3,330.41 for potential corporate/bulk clients.

Investigate the Second-Half Revenue Drop: determine why July–December revenue is consistently lower than January–June, particularly the September low of $69,321.65.

Optimize Marketing Spend: increase investment behind Instagram and Email, the top two referral sources by revenue, and assess whether direct Referral traffic (the lowest of the five) needs a refreshed incentive program.

Strategic Initiatives (Next 3–6 Months)

Launch a Low-Cost Segment Strategy: create a “Budget-Friendly” collection aimed at orders below the Q1 threshold ($410.52) to capture price-sensitive customers.

Leverage Coupon Codes: the dataset shows four active coupon types (FREESHIP, SAVE10, WINTER15, NO COUPON); compare order value and frequency across these to identify which incentive structure performs best before scaling further campaigns.

Complete the Payment Method Analysis: extend the pivot table to break out Credit Card, Debit Card, Gift Card, and Online alongside Cash, since this comparison is currently incomplete.

6. Conclusion

This EDA reveals a thriving but operationally challenged business. Chair and Printer lead on revenue, while Phone trails on both revenue and volume. The most pressing concern is fulfillment health: Cancelled and Returned orders (41.42%) outweigh the orders that were successfully Delivered or Shipped (38.83%). The dataset also shows a clear seasonal split, with revenue peaking in June and declining through the second half of the year rather than peaking in January. Addressing the cancellation/return rate, investigating the second-half revenue decline, and completing the payment method breakdown are the next priorities before this analysis is finalized.

Report Prepared By: Motunrayo Oyindamola Adegbayike

Contact: motunrayoadegbayike333@gmail.com